

Total No. of Questions : 8]

SEAT No. :

PC1965

[6353]-299

[Total No. of Pages : 2

**T.E. (Honors in Robotics)
PRINCIPLES OF ROBOTICS -I**

(2019 Pattern) (Semester-I) (304181 HR)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 Q.7, or Q.8.
- 2) Assume suitable data if necessary.
- 3) Draw neat sketches wherever necessary.

Q1) a) Explain working, principle and advantages of vacuum grippers. **[9]**

b) State the Characteristics of grippers. **[9]**

OR

Q2) a) Explain working and principle of Magnetic gripper. **[9]**

b) Classify grippers based on various criterias. **[9]**

Q3) a) Classify sensors based on working principle. **[8]**

b) Explain working principle and construction of capacitive sensors. **[9]**

OR

Q4) a) Sketch and explain LVDT. **[9]**

b) Explain working principle & construction of optical proximity sensors. **[8]**

Q5) a) Enlist steps in forward kinematic Analysis. **[9]**

b) Explain with neat sketch D-H. parameter **[9]**

OR

P.T.O.

- Q6)** a) State properties of generalised composite matrix. [9]
b) A 2 DOF planer RR manipulator has $L_1 = 120$ mm and $L_2 = 75$ mm. Determine joint angles using Geometric approach, so that face end is located at (100, 70). [9]

- Q7)** a) Explain Functional safety applications in Robotics. [8]
b) Explain various levels of Image processings. [9]

OR

- Q8)** a) Explain various applications of Robotics in Industry. [8]
b) Write short note on following: (Any Two) [9]
i) Pick & Place Robots
ii) Home automation
iii) Hospital & patient cases

